MILITARY SPECIFICATION

CASE, BRASS, CAL .30, NATIONAL MATCH

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, packaging, the fabrication, manufacture of one type of cartridge case designated as Case, Brass, Cal .30, National Match.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-636	-	Visual Inspection Standards for Small Arms Ammunition Through Cal .50
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein:

DRAWING

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKING DRAWINGS

7643674	-	Classification of Cartridge Case Defects, Small Arms Ammunition
8595432	-	Cartridge, Cal .30 Match, M72
8595433	-	Case, Brass

INSPECTION EQUIPMENT DRAWINGS

EL 8658201	-	Chamber Pressure
EL 8663615	-	Function and Casualty
7298354	-	Indicator Gage
7298478	-	Indicator Gage
7298485	-	Indicator Gage
7558058	-	Gage, Indicating, Diameter of Extractor Groove
8648308	-	Gage, Receiver, Profile and Alignment, Cal .30 Cartridge
8648315	-	Gage, Fixture, Indicating, Cal .30 Cartridge, Profile Failure
8648611	-	Lifting Lever Assembly
8648717	-	Instructions, Profile Failure Inspection

PUBLICATIONS

SCATP-7.62	-	Ammunition Ballistic Acceptance Test Methods, Test Procedures for 7.62MM Cartridge

2.1.3 Order of precedence. In the event of a conflict between this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The cartridge case shall comply with the requirements of Drawing (Dwg.) 8595433, referenced specifications and other requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

3.2 Function and casualty. The brass case shall be assembled in accordance with Dwg 8595432 and shall function without casualty when tested in the M1 rifle.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract. (See 4.3)

3.4 Workmanship. The brass case shall be fabricated in a thorough and workmanlike manner. It shall be free of burrs, sharp edges, foreign matter, and defects not permitted by MIL-STD-636. The cleaning method used shall not be injurious to the brass case, nor shall it be contaminated by the cleaning agents. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply are hereby made part of this detail specification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (See 4.3)

b. Quality conformance Inspection (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated.

Part Description	Drawing	Quantity
Cartridge, Cal .30 Match, M72	8555432	400
Case, Brass	8595433	125

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078 and Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

Table I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-70441(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Case, Brass				8595433
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case, Brass				
	Examination for defects	125	ACC 0 REJ 1	3.1, 3.4	4.4.2.1
	Function and Casualty	400	Table II	3.2	4.5.1 EL 8658201 EL 8663615

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot shall contain brass cases from one interfix lot number from one manufacturer. Inspection lots shall not exceed 150,000 cartridges cases per lot.

4.4.2 Examination. See MIL-A-48078.

Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70441(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Case, Brass	1 OF 3	8595433
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1.	Split case in I, J, K, L, M, or S location		100%	3.4	Visual 1/
2.	Perforated case		100%	3.4	Visual 1/
3.	Absence of vent hole or obstruction in vent hole area		100%	3.1	Visual
MAJOR:					
101.	Corrosion or stain		.25%	3.4	Visual 1/
102.	Round head		.25%	3.4	Visual 1/
103.	Dent		.25%	3.4	Visual 1/
104.	Draw scratch		.25%	3.4	Visual 1/
105.	Scratch		.25%	3.4	Visual 1/
106.	Beveled underside of head		.25%	3.4	Visual 1/
107.	Scaly metal		.25%	3.4	Visual 1/
108.	No chamfer on head (rim)		.25%	3.4	Visual 1/
109.	Fold, wrinkle or buckle		.25%	3.4	Visual 1/
110.	Bulge		.25%	3.4	Visual 1/
111.	Defective head		.25%	3.4	Visual 1/
112.	Defective mouth		.25%	3.4	Visual 1/

NOTES: 1/ Refer to MIL-STD-636 for visual defect standards. In the event of a conflict between paragraph 4.4.2.1 of this specification and MIL-STD-636 as to defect classification, the classification specified in paragraph 4.4.2.1 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70441(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 8595433
4.4.2.1	Case, Brass				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113.	Poor workmanship (excluding Criticals 1 and 2 and Majors 101 through 112)		.25%	3.4	Visual
114.	Diamter of primer pocket		.25%	3.1	SMTE
115.	Diameter of head		.25%	3.1	7298485
116.	Diameter of mouth		.25%	3.1	SMTE
117.	Height, overall		.25%	3.1	SMTE
118.	Length to shoulder datum		.25%	3.1	7298354
119.	Diameter of extractor groove		.25%	3.1	7558058
120.	Depth of primer pocket		.25%	3.1	SMTE
121.	Thickness of head		.25%	3.1	7298478
122.	Diameter vent hole		.25%	3.1	SMTE
123.	Thickness at mouth		.25%	3.1	SMTE
124.	Variation in mouth thickness		.25%	3.1	SMTE
125.	Wall thickness, at shortest distance, minimum		.25%	3.1	SMTE
126.	Wall thickness, at second shortest distance, minimum		.25%	3.1	SMTE
127.	Wall thickness at third shortest distance, minimum		.25%	3.1	SMTE
128.	Wall thickness at fourth shortest distance, minimum		.25%	3.1	SMTE
129.	Web thickness in vent hole area, minimum		.25%	3.1	SMTE

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.1	Case, Brass				8595433
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
130.	Profile and alignment		.25%	3.1	8648308, 8648315, 8648611, 8648717
131.	Weight		.25%	3.1	SMTE
132.	Function and casualty	240	Table II	3.2	4.5.1, EL 8658201, EL 8663615
MINOR:					
201.	Illegible or missing head stamp		1.5%	3.1	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

TABLE II.

Defects	Acceptance Number 1/ 1st Sample	Cummulative (1st & 2nd Samples)
1. Misfire 2/		
(a) No vent hole or obstruction in the vent hole area	0	-
(b) Other	1	2
2. Primer leak 5:		
(a) Perforation in firing pin indent in primer cup	1	2
(b) Escape of gas through primer cup other than (a) above	1	2
(c) Escape of gas around primer cup3.	5	9
(d) Blown primer or primer falls out of pocket on retraction of bolt	0	1
(e) Primer remains in pocket but is physically loose	1	2
3. Case casualties:		
(a) Longitudinal split4:		
(1) Neck and shoulder (I&S)	9	17
(2) Body (J)	3	5
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
(b) Circumferential rupture4:		
(1) Partial, shoulder or body (J or S)	2	4
(2) Partial body (K)	0	1
(3) Partial head (L)	0	1
(4) Complete	0	1
4. Failure to extract	0	1

1 If the number of defects found in the first sample exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantity specified under the function and casualty test, shall be fired in the weapon If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the first article sample or lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the first article sample or lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

[2] Each cartridge that misfires shall be disassembled and examined for the absence of the vent hole in the primer pocket or any obstruction in the vent hole area of the primer pocket that can be assignable as the cause for misfire. If the vent hole is missing or obstructed, the lot shall be rejected with no second sample permitted.
[3] Gas escapes around more than 50 percent of periphery of cup.
[4] For location of defects indicated by letters in parentheses, see Dwg. 7643674.
[5] If a lot fails due to primer leak an investigation will be made by the contractor to determine if the case or the primer was the cause of failure. Final determination of the cause of failure will be determined by the government.

4.4.3 Testing. Testing is described in Table 1 and paragraph 4.4.2.1.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in Table 1 and paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Function and casualty test. Prior to assembly of this sample, a quantity of 20 cartridges, assembled in accordance with Dwg. 8595432, shall be tested for chamber pressure in accordance with Section 7 of SCATP-7.62. The test shall be conducted at $70^{\circ} \pm 2^{\circ}F$. If any individual chamber pressure reading is not between 47,500 pounds per square inch (psi) plus or minus 2500 psi, a new charge weight shall be established. When the individual pressure readings from a quantity of 20 cartridges meet the pressure requirement, the function and casualty sample shall be assembled in accordance with Dwg. 8595432 using the assessed charge weight. The charge weight and overall length of each of the assembled cartridges from both the assessment quantity and function and casualty sample shall be measured and the values recorded. The function and casualty test shall be performed in accordance with Section 10 of SCATP-7.62 using two M1 rifles.

5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified, packaging shall be Level B. Maximum and intermediate protection (Level A and B) shall be in accordance with MIL-STD-1169. Minimum protection shall be in accordance with MIL-STD-1188.

5.2 Unitization. Unitization for Levels A and B shall be in accordance with MIL-STD-1169.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The cartridge case covered by this specification is intended for National Match use.

6.2 Ordering data.

a. See MIL-A-48078

b. Responsibility for furnishing test weapons, bullets, propellant and primer (See 3.2).

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDC, ATTN: DRSMC-QAF-I(D), Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as U.S. Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-A935)

☆U.S. GOVERNMENT PRINTING OFFICE: 1984—705-040/A0710

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT RESEARCH & DEVELOPMENT CENTER
ATTN: DRSMC-QA
DOVER, NJ 07801

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70441	CASE, BRASS, CAL .30, NATIONAL MATCH

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T03:26Z
Check the source to verify that this is the current version before use.